

Rebuttal for CoRL Submission #18

We thank the reviewers and Area Chair for identifying important ambiguities in the optimization, masking, ablations, and real-robot protocol. We agree that two implementation choices require correction, and that the component and human-held evidence must be separated more carefully. Below we report the currently auditable results and mark the missing controlled comparisons.

TABLE I

Operator-reported, operator-held SWIPE results for isolated changes to Legacy Full (success is one internal roller rotation within 60 s). CIs are 95% Wilson intervals; the aligned Legacy Full anchor is not yet available.

Isolated change from Legacy Full	Success/ N	95% CI
None (same-protocol Legacy Full)	TBD	TBD
Learned loss weight $\rightarrow$ fixed $1 + p$	10/10	[0.722, 1.000]
Whole-lowdim $\rightarrow$ tactile-only mask	13/20	[0.433, 0.819]
Unbounded $\rightarrow$ bounded gate	7/20	[0.181, 0.567]

These isolated retrainings change one mechanism at a time. The fixed Alpha1 weight is the strongest current variant; an exploratory two-sided Fisher test against the bounded-gate variant gives $p=0.0011$. We do not claim improvement over Legacy Full until its same-protocol $N=20$ anchor is complete. We also exclude the earlier $N=5$ hand-held pilots and unlabelled rollout files from this table rather than pooling incompatible protocols.

Q2–Q3: Loss weighting and proprioceptive masking. We agree that a learned scalar optimized through the same weighted denoising objective can be driven to its lower bound. We replace it with the fixed, bounded schedule $w(c) = \text{clip}(1 + c, 1, 2)$. Its 10/10 preliminary result supports retaining this correction and extending it to $N=20$. We also agree that the original post-encoder whole-lowdim mask was too coarse. The corrected routing always

retains proprioception and gripper width and masks only tactile features. Its 13/20 result establishes usability, but not superiority to the pending Legacy Full anchor. These are implementation corrections rather than a defense of the original formulations.

Q4: Is feature gating necessary? The bounded-gate variant obtains 7/20, and qualitative traces show reluctance to advance unless a strong visual cue is introduced. Thus the tested gate range appears to over-suppress vision. This result rejects that parameterization, not gating in general. We will soften the independent gating claim, select one weaker setting using offline prediction/action traces, freeze it before robot evaluation, and report the result even if it is negative.

Q1: Is VPCE only task progress or phase? We implemented same-backbone Phase-DP and Progress-DP baselines, but found that their real-time conditioning must be aligned to actual elapsed/control time; therefore the existing pilot and unlabelled rerun files are not used as formal evidence. We will report corrected fixed-mount $N=20$ comparisons. Until then, we narrow the claim: VPCE predicts a tactile-supervised future-contact signal online from RGB and motion, but superiority to explicit progress remains TBD.

Human-held props and limitations. The current table is explicitly operator-held; it cannot resolve the compliance concern. Our minimum controlled study rigidly mounts the reader and randomizes $N=20$ trials each for DP-VT, Legacy Full, and the best fixed-Alpha1 variant under identical initialization, controller, and timeout. Only one complete roller rotation counts as success. We will also report task-specific label calibration and target adaptation, and do not claim zero-shot transfer to unseen tasks, objects, or viewpoints.